

- b) $\{(1, 2), (1, 5), (1, 7), (1, 8), (2, 5), (2, 7), (2, 8), (5, 7), (5, 8), (7, 8)\}$
 c) $\{(1, 5), (2, 7)\}$
 d) $\{(5, 1)\}$

23. Se $A = \{1, 2, 3, 4, 5, 6\}$, o conjunto dos resultados possíveis será:

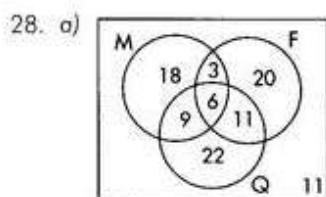
$A \times A = A^2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$.

24. $\{(K, K), (K, C), (C, K), (C, C)\}$, em que K e C representam cara e coroa, respectivamente.

25. b) $E - A$ c) E d) A

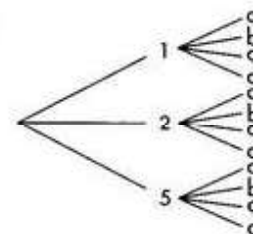
26. $n = 10$

27. a) $A_1 = \{1\}; A_2 = \{2\}; A_3 = \{3, 4, 5, 6\}$
 b) $A_1 = \{1, 2\}; A_2 = \{3, 4\}; A_3 = \{5, 6\}$



- b) 29
 c) 36, 40 e 48
 d) 64 e 89

29.



30. 1: $A \cap B \cap C$

2: $A \cap C - A \cap B \cap C$

3: $A \cap B - A \cap B \cap C$

4: $B \cap C - A \cap B \cap C$

5: $A - (A \cap B) \cup (A \cap C)$

6: $B - (A \cap B) \cup (B \cap C)$

7: $C - (A \cap C) \cup (B \cap C)$

31. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$

32. 170

33. a) 2% b) 41%

34. Não

Capítulo 2

1. a) F f) F
 b) F g) V
 c) F h) V
 d) F i) F
 e) V j) F

2. a) 0,4 d) 0,32
 b) 1,666... e) 0,2525...
 c) 1,4 f) 0,4666...

3. a) 1,28 d) 1,07
 b) 0,28 e) 1,52
 c) 0,63 f) 0,15

4. a) $\frac{43}{100}$ c) $\frac{2.454}{1.000}$ e) $-\frac{72}{100}$
 b) $\frac{7}{100}$ d) $\frac{1.212}{100}$ f) $\frac{31.415}{10.000}$

5. a) $\frac{8}{9}$ c) $\frac{23}{9}$ e) $\frac{59}{90}$
 b) $\frac{24}{99}$ d) $\frac{65}{90}$ f) $\frac{563}{900}$

6. $x=0$ e $y=0$

7. a) 3,4641 b) 5,4772 c) 8,8318 d) 22,3607

8. a) $S = \{16\}$ e) $S = \left\{\frac{17}{3}\right\}$ h) $S = \{1, 5\}$
 b) $S = \{7\}$ f) $S = \{0\}$ i) $S = \{14\}$
 c) $S = \{3\}$ g) $S = \{0\}$ j) $S = \left\{\frac{81}{7}\right\}$
 d) $S = \{3\}$

9. a) $S = \left\{\frac{5}{7}\right\}$ h) $S = \left\{\frac{-6}{5}\right\}$
 b) $S = \left\{\frac{67}{8}\right\}$ i) $S = \{-2\}$
 c) $S = \left\{\frac{14}{5}\right\}$ j) $S = \{-10\}$
 d) $S = \{-1\}$ k) $S = \left\{\frac{33}{14}\right\}$
 e) $S = \left\{\frac{36}{53}\right\}$ l) $S = \left\{\frac{M-100}{100}\right\}$
 f) $S = \left\{\frac{28}{15}\right\}$ m) $S = \left\{\frac{18K-29}{2}\right\}$
 g) $S = \left\{\frac{31}{44}\right\}$ n) $S = \left\{\frac{1+3y}{y-2}\right\}, y \neq 2$

10. 140 unidades

11. 200 camisas

12. 50 meses

13. a) $S = \{x \in \mathbb{R} | x > 5\}$
 b) $S = \{x \in \mathbb{R} | x > -4\}$
 c) $S = \{x \in \mathbb{R} | x \geq -6\}$
 d) $S = \{x \in \mathbb{R} | x \leq 0\}$
 e) $S = \left\{x \in \mathbb{R} | x > \frac{5}{3}\right\}$
 f) $S = \left\{x \in \mathbb{R} | x \geq \frac{27}{5}\right\}$
 g) $S = \{x \in \mathbb{R} | x \leq -7\}$
 h) $S = \left\{y \in \mathbb{R} | y \geq \frac{43}{11}\right\}$
 i) $S = \left\{m \in \mathbb{R} | m \leq \frac{5}{2}\right\}$

14. 500 unidades

15. entre 180 e 380 unidades

16. a) $S = \{1, 4\}$ g) $S = \{0, 5\}$
 b) $S = \{3, 4\}$ h) $S = \{3 + 2\sqrt{3}, 3 - 2\sqrt{3}\}$
 c) $S = \{2, 4\}$ i) $S = \{1 + \sqrt{6}, 1 - \sqrt{6}\}$
 d) $S = \{2\}$ j) $S = \emptyset$
 e) $S = \emptyset$ k) $S = \{2, -6\}$
 f) $S = \{1, 2\}$ l) $S = \emptyset$

17. a) $S = \{0, 5\}$ d) $S = \{4, -4\}$
 b) $S = \{0, 3\}$ e) $S = \{2, -2\}$
 c) $S = \{5, -5\}$ f) $S = \{0\}$

18. $-9/2$

19. $\frac{3+\sqrt{5}}{2}$ ou $\frac{3-\sqrt{5}}{2}$

20. $x=2$ ou $x=8$

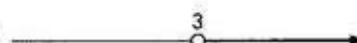
21. $x=5$

22. \$ 5,00 ou \$ 15,00

23. a) $[2, 20]$ c) $[2, 7[$
 b) $[7, 8]$ d) $]-\infty, 2[\cup]8, \infty[$

24. a) $[1, 5[$ b) $[0, \infty[$ c) $[5, \infty[$

25. a) 

b) 

c) 

d) 

e) 

f) 

g) 

h) 

26. a) $-12 < x < 12$ d) $x > 8$ ou $x < -8$
 b) $3 < x < 9$ e) $x > 9$ ou $x < 5$
 c) $4 > x > -3$ f) $x < -1$ ou $x > 7/3$

27. $16,16 < x < 31,84$

28. $10,05 < x < 19,95$

Capítulo 3

1. a) $\{(1, 3), (1, 5), (1, 8), (1, 9), (3, 5), (3, 8), (3, 9), (5, 8), (5, 9), (7, 8), (7, 9)\}$
 b) $\{(3, 3), (5, 3), (5, 5), (7, 3), (7, 5)\}$
 c) $\{(1, 3), (1, 5), (1, 8), (1, 9), (3, 3), (3, 9), (5, 5)\}$
 d) $\{(3, 3), (5, 5)\}$
 e) $\{(1, 3), (3, 5), (7, 9)\}$

2. diagrama de flechas da resposta anterior.

3. a) $D = \{1, 3, 5, 7\}$, $Im = \{3, 5, 8, 9\}$
 b) $D = \{3, 5, 7\}$, $Im = \{3, 5\}$
 c) $D = \{1, 3, 5\}$, $Im = \{3, 5, 8, 9\}$
 d) $D = \{3, 5\}$, $Im = \{3, 5\}$
 e) $D = \{1, 3, 7\}$, $Im = \{3, 5, 9\}$

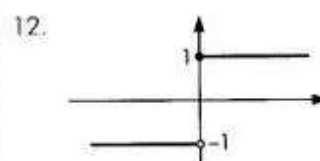
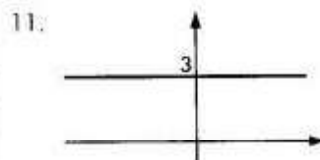
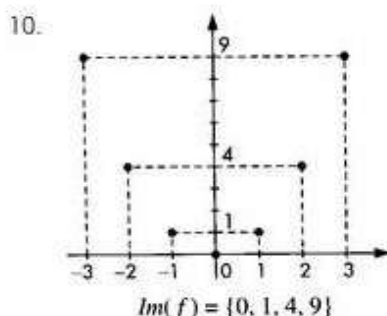
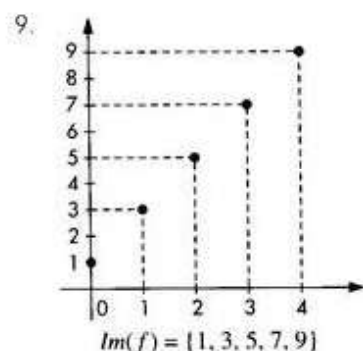
4. a) 11
 b) 39
 c) -3
 d) -10
 e) $7\sqrt{2} - 3$
 f) $1/2$
 g) $-16/3$
 h) $7(a + b) - 3$

5. a) 3
 b) -11
 c) 26
 d) $-7/2$

6. a) x_0^2
 b) $(x_0 + h)^2$
 c) $2x_0h + h^2$

7. 1 e 3

8. $m = 3$



13. a) $R(x) = 5x$
 b) \$ 200,00
 c) 140 unidades

14. a) \$ 120,00
 b) \$ 2,00

15. a) 4.333,33
 b) 234,33

16. a) \$ 29,00
 b) 16,50
 c) \$ 4,00

17. a) \$ 60,00
 b) \$ 150,00
 c) $\begin{cases} y = 0,1x & \text{para } x \leq 900 \\ y = 90 + 0,2(x - 900) & \text{para } x \geq 900 \end{cases}$

18. a) \$ 8,00
 b) \$ 13,00
 c) \$ 41,80
 d) $\begin{cases} y = 8 & \text{para } x \leq 10 \\ y = 8 + 1(x - 10) & \text{para } 10 \leq x \leq 20 \\ y = 18 + 1,4(x - 20) & \text{para } x \geq 20 \end{cases}$

19. $S(x) = 2000 + 50x$

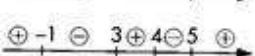
20. $A = x \cdot (20 - x)$

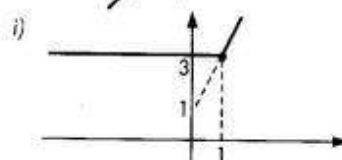
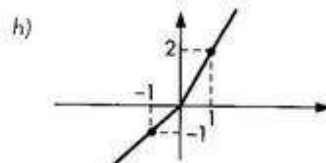
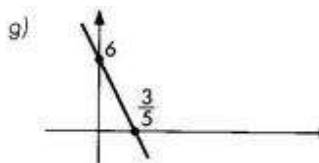
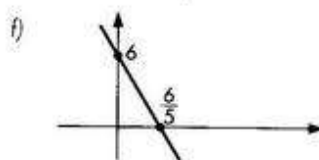
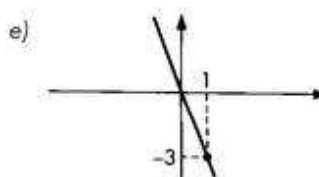
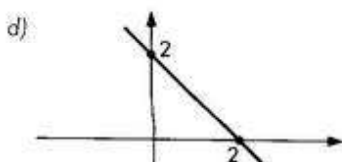
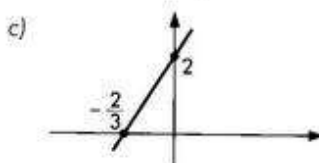
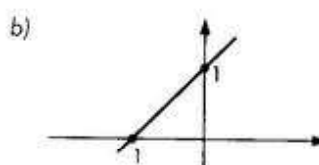
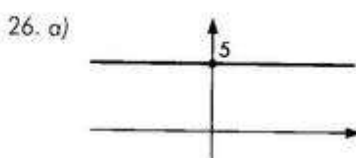
21. $A = \frac{\sqrt{3}}{4} \cdot x^2$

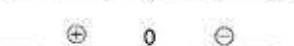
22. São funções: a, c, d, f, i, j, l.

23. a) R
 b) $R - \{2\}$
 c) $R - \{0, 3\}$
 d) $[0, \infty[$
 e) $[2, \infty[$
 f) $] -\infty, 2]$
 g) $]1, \infty[$
 h) $[3, \infty[$
 i) $[3, \infty[$
 j) $[2, \infty[$

24. Crescente: $[-7, -4]$, $[-1, 6]$
 Decrescente: $[-4, -1]$, $[6, 7]$
 Pontos de máximo: -4 e 6
 Pontos de mínimo: -7 , -1 e 7

25. a) 
 b) 
 c) 
 d) 
 e) 



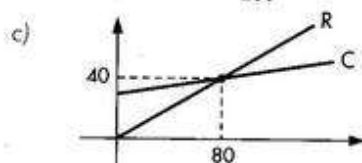
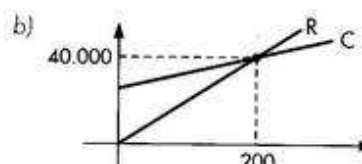
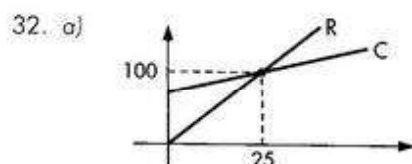
27. a) 
 b) 
 c) 
 d) 
 e) 

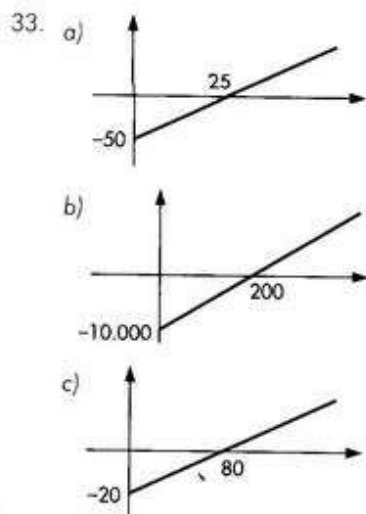
28. a) 5
 b) 1
 c) $1/4$
 d) $-3/7$

29. a) $y = 2x + 1$
 b) $y = 3x$
 c) $y = -x + 3$
 d) $y = 2x$
 e) $y = -3x - 4$
 f) $y = -x - 2$

30. a) $y = x + 1$
 b) $y = \frac{2}{5}x + \frac{2}{5}$
 c) $y = -\frac{3}{2}x + 4$

31. a) $y = -\frac{3}{4}x + 3$
 b) $y = \frac{1}{2}x + 2$
 c) $y = -x + 5$





34. 500 unidades
35. 900 unidades
36. a) 500 unidades
b) 400 unidades
c) 75%
37. a) $L = 2x - 30\,000$
b) $LL = 1,4x - 21\,000$
38. 807,7 unidades
39. a) $R = 10x$
b) $C = 150 + 7x$
c) 50
d) $L = 3x - 150$
e) 110
40. a) 227,3 unidades
b) \$ 11,00
c) \$ 8.500,00
d) 64,7%
41. a) \$ 200,00
b) 300/7
42. \$ 27,50
43. 500 unidades
44. Não é vantajosa.
45. 2 horas
46. 100 km
47. a) $C = 2.000 + 24x$
b) \$ 9.200,00
48. $C = 6.000 + 60x$
49. \$ 25,00
50. a) $p = 1,25c$
b) 25%
51. 42,86%

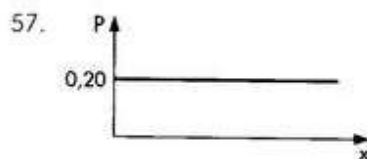
52. 20%

53. Demonstração.

$$54. m_p = \frac{m_c}{1 + m_c}$$

$$55. p = -0,2x + 30$$

$$56. p = -0,01x + 7$$



$$58. p = 0,004x - 10$$

$$59. p = 300 + 0,5 \cdot x$$

60. a) demanda
b) oferta
c) oferta
d) demanda
e) oferta

61. a) \$ 15,00
b) \$ 42,50

62. a) \$ 2,00
b) 600 ton
c) $x = 650$ ton e $p = 3,5$

63. a) \$ 14,00
b) 25 unidades
c) \$ 12,00

64. 182,2

65. 20 unidades

66. a) $p = 60 - 2x$; $p = 70 - 2x$; $p = 80 - 2x$.
b) Desloca-se paralelamente para cima.

67. a) $p = 41 + 0,5x$
b) $p = 39 + 0,5x$

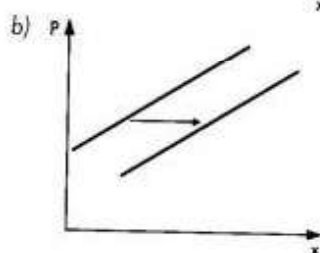
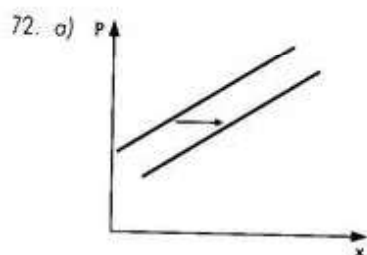
68. a) \$ 70,00
b) \$ 73,00
c) \$ 162,00

69. \$ 58,00

70. a) \$ 40,00
b) \$ 43,38

Sugestão: faça, por exemplo, $p = 100$ na função oferta; o valor de x será 160. Portanto, na nova curva de oferta, teremos $p = 110$ (10% a mais) e $x = 160$. Obtenha, de modo análogo, outro ponto da curva de oferta e ache a equação da reta de oferta.

71. a) \$ 40,00
b) \$ 46,67



73. a) \$ 1.400,00

b) \$ 600,00

c) 10 anos

74. a) \$ 6.000,00

b) \$ 3.500,00

75. \$ 18.333,33

76. a) $V = 10.000 - \frac{4.000}{3} \cdot x$ b) $\frac{16.000}{3}$

77. 7,5 anos

78. a) $C = 800 + 0,8y$

b) $S = -800 + 0,2y$

79. a) $S = 0,4y - 500$

b) \$ 1.250,00

80. a) $C = 800 + 0,65y$

b) \$ 1.000,00

81. \$ 500,00

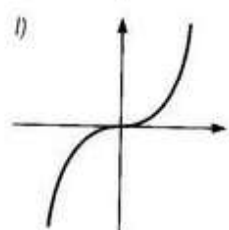
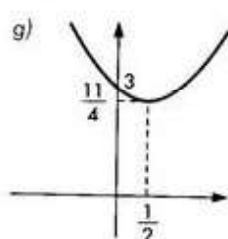
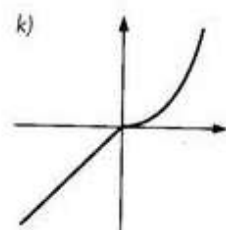
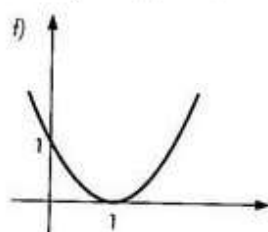
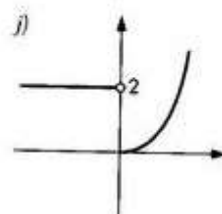
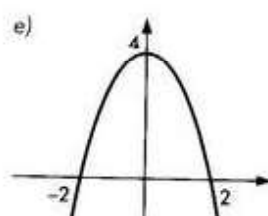
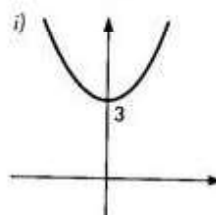
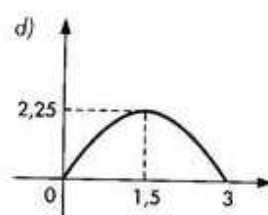
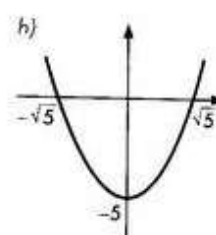
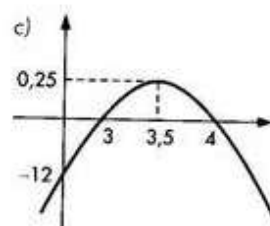
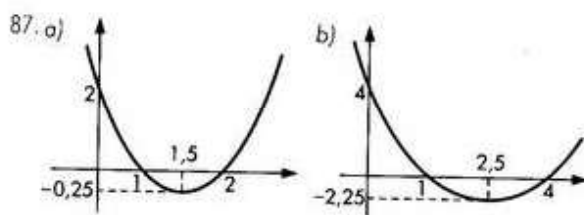
82. \$ 750,00

83. \$ 60,00

84. \$ 85,00

85. $C = 0,6y + 2.000$

86. $S = 0,4y - 2.000$



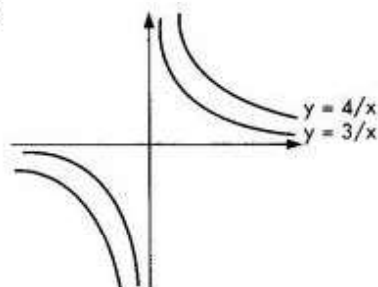
88. a) $\oplus \quad 1 \quad \ominus \quad 2 \quad \oplus$
Ponto de mínimo: $x = 1,5$
Conjunto-imagem: $[-0,25, \infty[$

b) $\oplus \quad 1 \quad \ominus \quad 4 \quad \oplus$
Ponto de mínimo: $x = 2,5$
Conjunto-imagem: $[-2,25, \infty[$

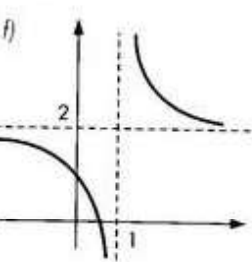
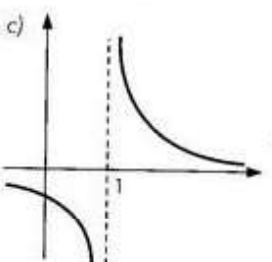
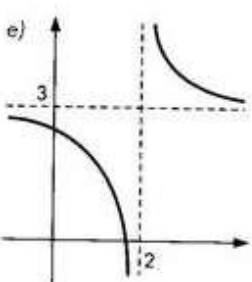
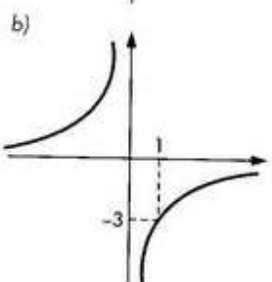
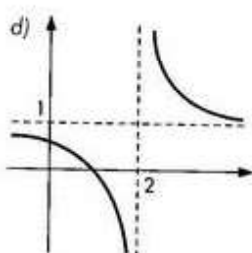
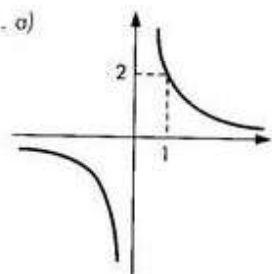
c) $\ominus \quad 3 \quad \oplus \quad 4 \quad \ominus$
Ponto de máximo: $x = 3,5$
Conjunto-imagem: $]-\infty, 0,25]$

d) $\ominus \quad 0 \quad \oplus \quad 3 \quad \ominus$
Ponto de máximo: $x = 1,5$
Conjunto-imagem: $]-\infty, 2,25]$

- e) $\ominus \quad -2 \quad \oplus \quad 2 \quad \ominus$
 Ponto de máximo: $x = 0$
 Conjunto-imagem: $]-\infty, 4]$
- f) $\oplus \quad 1 \quad \oplus$
 Ponto de mínimo: $x = 1$
 Conjunto-imagem: $[0, \infty[$
- g) $\oplus$
 Ponto de mínimo: $x = 1/2$
 Conjunto-imagem: $\left[\frac{11}{4}, \infty\right[$
- h) $\oplus \quad -\sqrt{5} \quad \ominus \quad \sqrt{5} \quad \oplus$
 Ponto de mínimo: $x = 0$
 Conjunto-imagem: $[-5, \infty[$
- i) $\oplus$
 Ponto de mínimo: $x = 0$
 Conjunto-imagem: $[3, \infty[$
- j) $\oplus \quad 0 \quad \oplus$
 Ponto de mínimo: $x = 0$
 Conjunto-imagem: $[0, \infty[$
- k) $\ominus \quad 0 \quad \oplus$
 Conjunto-imagem: $\mathbb{R}$
- l) $\ominus \quad 0 \quad \oplus$
 Conjunto-imagem: $\mathbb{R}$
89. a) $y > 0$ para $1 < x < 3$ ou $x > 5$
 $y < 0$ para $x < 1$ ou $3 < x < 5$
 $y = 0$ para $x = 1$ ou $x = 5$
 b) $y > 0$ para $x < -2$ ou $2 < x < 3$
 $y < 0$ para $-2 < x < 2$ ou $y > 3$
 $y = 0$ para $x = 3$
 c) $y > 0$ para $x < -1$
 ou $0 < x < 1$ ou $x > 3$
 $y < 0$ para $1 < x < 3$ ou $-1 < x < 0$
 $y = 0$ para $x = 1$ ou $x = -1$
 d) $y > 0$ para $-2 < x < 2$ ou $x > 4$
 $y < 0$ para $x < -2$ ou $2 < x < 4$
 $y = 0$ para $x = 2$ ou $x = 4$
90. a) $\{x \in \mathbb{R} \mid x \leq 0 \text{ ou } x \geq 6\}$
 b) $\{x \in \mathbb{R} \mid 0 \leq x \leq 3\}$
 c) $\{x \in \mathbb{R} \mid x < -2 \text{ ou } x > 2\}$
 d) $\{x \in \mathbb{R} \mid 0 < x \leq 3 \text{ ou } x > 6\}$
 e) $\{x \in \mathbb{R} \mid x < -4 \text{ ou } -1 \leq x \leq 1\}$
91. a) Ponto de máximo: $x = 2$
 Ponto de mínimo: $x = 4$
- b) Ponto de máximo: $x = 2$
 Ponto de mínimo: $x = 0$
- c) Pontos de máximo: $x = -1$ ou $x = 1$
 Ponto de mínimo: $x = 0$
- d) Ponto de máximo: $x = 5$
 Ponto de mínimo: $x = 8$
92. a) $x = 5$ b) $x = 19/4$
93. a) $x = 20$ b) $x = 4,5$
94. a) $p = -0,02x + 40$ b) \$ 30,00
95. a) $p = -0,05x + 25$ b) \$ 12,50
96. a) $p = -0,015x + 13$ b) \$ 8,50
97. a) \$ 7,50 b) \$ 9,00
98. a) $p = -5x + 350$ b) \$ 175,00
99. a) \$ 60,00 b) \$ 50,00
100. a) \$ 51,00 c) \$ 40,00
 b) \$ 51,00
101. a) $R = -0,01x^2 + 10x$, $0 \leq x \leq 600$
 b) \$ 5,00
102. a) $p = -0,02x + 8$ b) \$ 4,00
103. a) \$ 51,50 b) \$ 52,50
104. a) \$ 19,00 b) \$ 12,50
105. a) \$ 21,50 b) \$ 10,00
106. a) $L = -3x^2 + 180x - 200$
 b) $x = 30$
107. 80/3
108. a) \$ 6,50 b) $6 \leq p \leq 7$
109. 3,5 unidades
- 110.



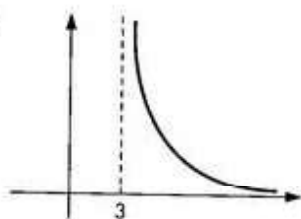
111. a)



112. a) $x=1$ e $p=10$
 b) $x=1$ e $p=8$
 c) $x=1$ e $p=55/7$

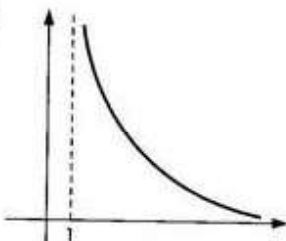
113. a) $C = \frac{12 \cdot 10^5}{x}$ b) \$ 3.000,00

114. a)



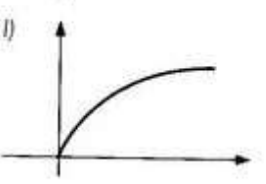
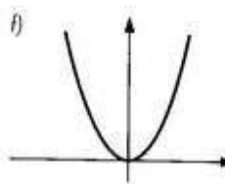
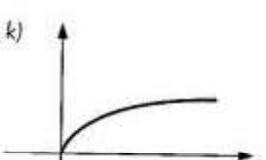
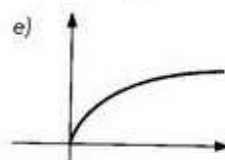
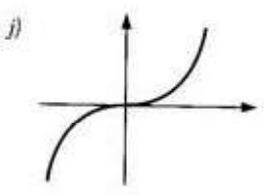
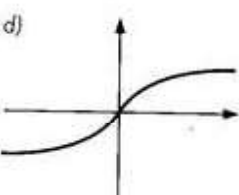
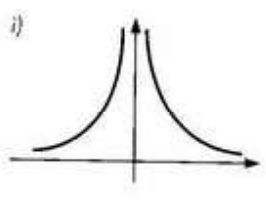
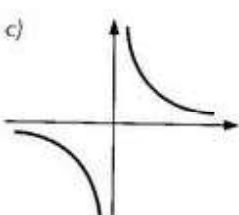
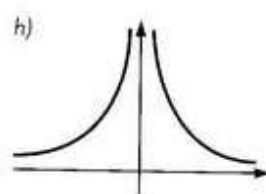
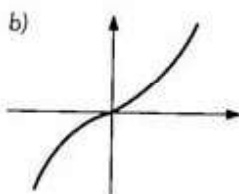
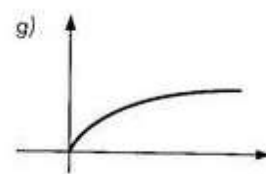
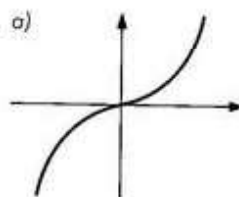
- b) 2,5 bilhões c) tende a infinito

115. a)



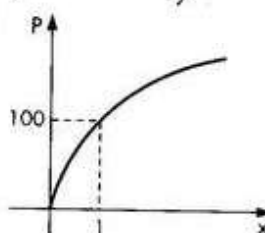
- b) 2,5 bilhões c) tende a infinito

116. a)

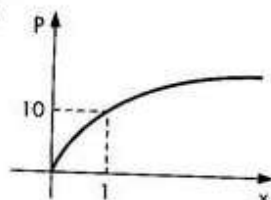


117. a) $x=1$ e $p=1$ b) $x=1$ e $p=1$

118. a) 400; 25
 b) 800; 12,5
 c) Dobrará.
 d) 0
 e)



119. a) 40; 5
b) 160; 2,5
c) Quadruplicará.
d) 0
e)



120. a) 19 b) 48 c) \$ 1.698,50

121. a) 63 b) \$ 1.294,96 c) \$ 599,48

122. a) 16 f) 1/16 j) 4/9
b) 81 g) 1/25 k) -8/27
c) 1 h) 1/8 l) 3/2
d) 1/9 i) 1/81 m) 9/4
e) 1/8

123. a) x^5 d) x^7y^7 g) 4
b) x^9 e) $144x^6$ h) 1/2
c) x^4 f) 2 i) $1+i$

124. a) 2 d) 2,8284 g) 1,0188
b) 5 e) 14,6969 h) 1,8286
c) 1,7321 f) 2,5119

125. a) 16 c) 3 e) 0,25
b) 6 d) 128

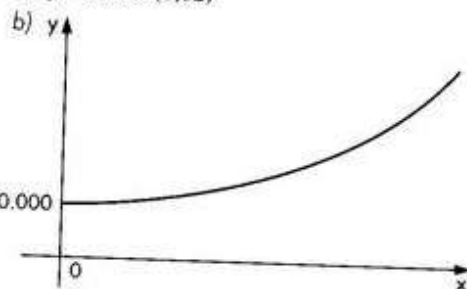
126. a) 8.867 b) 16.990

127. 3,53%

128. 2,81%

129. 765,77 bilhões de dólares

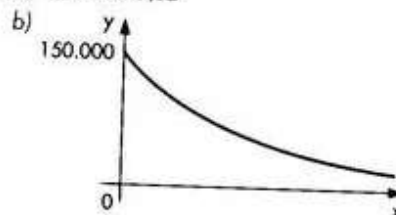
130. a) $y = 20.000(1,02)^x$



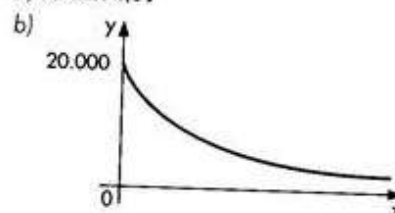
131. 25.598 habitantes

132. 2.488,32 unidades

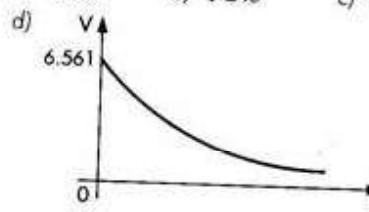
133. a) \$ 110.613,62



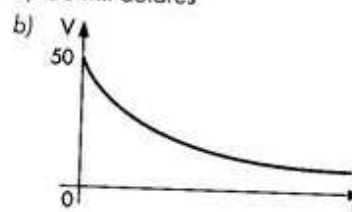
134. a) \$ 8.874,11



135. a) 6.561 b) \$ 243 c) 6.318



136. a) 50 mil dólares

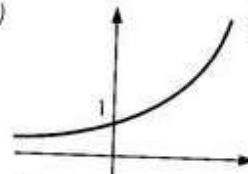


137. $K = -2,60\%$

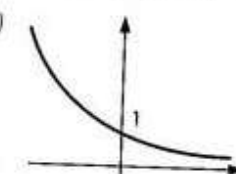
138. $V = 4.000 (0,866)^t$

139. a) \$ 25.000 b) $17.000 \cdot (0,9)^{x-2}$ para $x \geq 2$

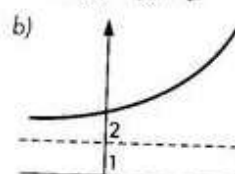
140. a)



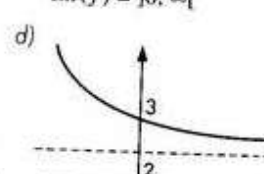
$D = R$
 $Im(f) =]0, \infty[$



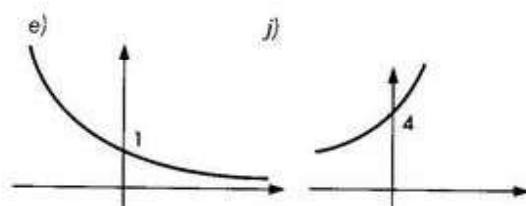
$D = R$
 $Im(f) =]0, \infty[$



$D = R$
 $Im(f) =]1, \infty[$

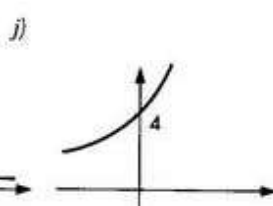


$D = R$
 $Im(f) =]2, \infty[$



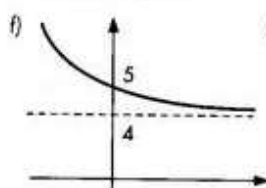
$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$



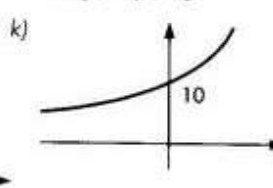
$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$



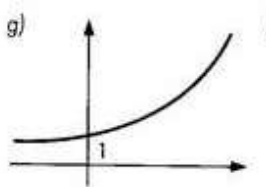
$$D = \mathbb{R}$$

$$\text{Im}(f) =]4, \infty[$$



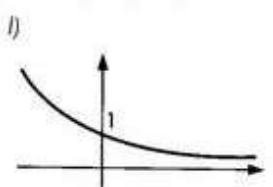
$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$



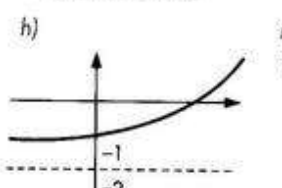
$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$



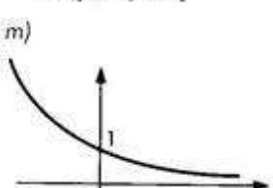
$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$



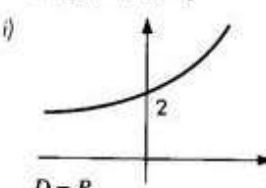
$$D = \mathbb{R}$$

$$\text{Im}(f) =]-2, \infty[$$



$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$



$$D = \mathbb{R}$$

$$\text{Im}(f) =]0, \infty[$$

141. a) 3 e) 1 i) -2
 b) 2 f) 4 j) -1/2
 c) 4 g) -3 k) 6
 d) 0 h) -2 l) 6

142. a) 1,7324 g) 0,4055 m) 2,8074
 b) 0,8451 h) 5,4931 n) 1,2789
 c) 2,0864 i) 0,5306 o) 0,5681
 d) 1,5391 j) -0,2231
 e) 3,4340 k) -0,0834
 f) 1,9459 l) -0,6162

143. a) 0,78 f) 2,48
 b) 0,90 g) 0,70
 c) 1,08 h) 1,70
 d) 1,38 i) -0,70
 e) 1,30 j) -1,52

144. a) $x = 0,625$ d) $x = 1,15$
 b) $x = 0,80$ e) $x = 1,67$
 c) $x = 3,20$ f) $x = -0,87$

145. a) -3,9694 d) 3,4650
 b) 2,5779 e) 7,8074
 c) 1,4882

146. a) 1,3863 c) 0,4241
 b) 0,8214 d) 0,1329

147. 23,5 anos aproximadamente

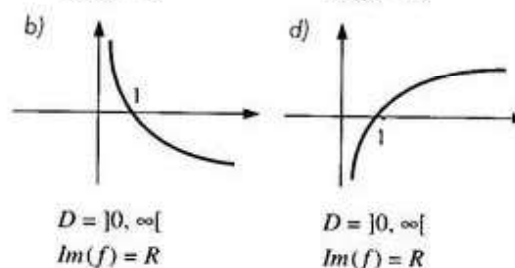
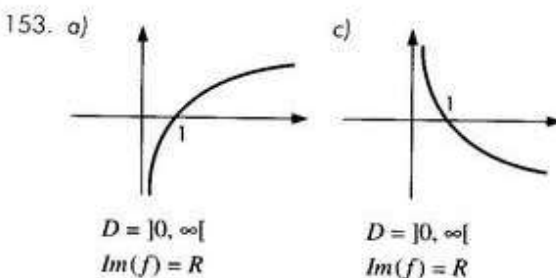
148. 22,5 anos

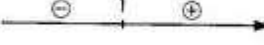
149. 22,7 anos aproximadamente

150. 4,3 anos aproximadamente

151. 3,1 anos aproximadamente

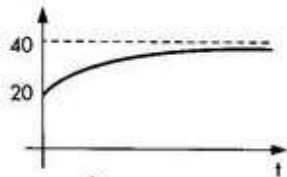
152. 15 anos



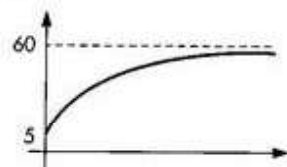
154. a) 
- b) 
- c) 
- d) 

155. a) $\{x \in \mathbb{R} | x > 3\}$
 b) $\{x \in \mathbb{R} | x < 2\}$
 c) $\{x \in \mathbb{R} | x < 1 \text{ ou } x > 3\}$
 d) $\{x \in \mathbb{R} | x < -2 \text{ ou } x > 2\}$
 e) $\{x \in \mathbb{R} | 0 < x < 4\}$

156. a) 20
 b) 32,6
 c) 40
 d)



157. a) 5
 b) 52,3
 c) 60
 d)



158. $K = 0,22$ e $B = 6,25$

159. \$ 2.147,93

160. \$ 14.282,46

161. \$ 3.313,47

162. \$ 3.814,48

163. 3,71% ao mês

164. 36,8 meses aproximadamente

165. 1,6 anos

166. \$ 5.327,83

167. \$ 8.556,29

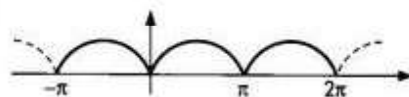
168. 10% ao mês

169. 8% ao mês

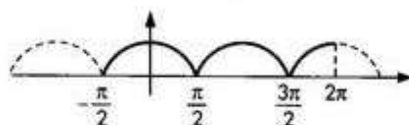
170. 5% ao mês

171. a) R\$ 279.161,54 b) R\$ 277,91

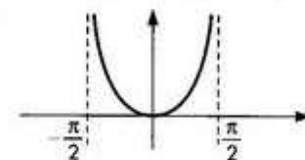
172. a)



b)



c)



173. a) $0 \quad \oplus \quad \pi \quad \ominus \quad 2\pi$

b) $0 \quad \oplus \quad \pi/2 \quad \ominus \quad 3\pi/2 \quad \oplus \quad 2\pi$

c) $-\pi/2 \quad \ominus \quad 0 \quad \oplus \quad \pi/2$

d) $0 \quad \ominus \quad 1 \quad \oplus \quad \pi \quad \ominus \quad 2\pi$

e) $0 \quad \ominus \quad \pi/2 \quad \oplus \quad 3 \quad \ominus \quad 3\pi/2 \quad \oplus \quad 2\pi$

f) $-\pi/2 \quad \ominus \quad -1 \quad \oplus \quad 0 \quad \ominus \quad 1 \quad \oplus \quad \pi/2$

174. a) Ponto de máximo: $\pi/2$

Pontos de mínimo: 0 e π

b) Ponto de máximo: 2π

Ponto mínimo: π

c) Ponto de mínimo: 0

d) Pontos de mínimo: $0, \pi$ e 2π

Pontos de máximo: $\pi/2$ e $3\pi/2$

e) Pontos de mínimo: $\pi/2$ e $3\pi/2$

Pontos de máximo: $0, \pi$ e 2π

Capítulo 4

1. a) $f(n) = n^2$ (divergente)
 b) $f(n) = (-1)^n \cdot n$ (divergente)
 c) $f(n) = 2^{n-1}$ (divergente)
 d) $f(n) = 5(n-1)$ (divergente)

e) $f(n) = \left(\frac{1}{3}\right)^{n-1}$ (converge para 0)

f) $f(n) = (0,1)^n$ (converge para 0)